

11_DOCUMENTATION.md

TASK 11

DOCUMENTATION & FINAL REPORT

Status

Mandatory

Owner

Technical Writer

Supporting Guides

Project Director

Requirements Engineer

Security Architect

Solution Architect

Database Architect

UI/UX Lead

Backend Lead

Frontend Lead

QA & Security Lead

Formal Methods Engineer

Principal Engineer

Estimated Time

3 to 5 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

11_TECHNICAL_WRITER.md

02_REQUIREMENTS.md

03_ARCHITECTURE.md

04_DATABASE.md

05_UI_DESIGN.md

06_BACKEND.md

07_FRONTEND.md

08_SECURITY.md

09_TESTING.md

10_FORMAL_METHODS.md

Mission

Produce complete, professional and distinction-level documentation for the Secure Dance Academy Management System.

Every engineering decision must be documented.

Every major feature must be explained.

Every diagram must support understanding.

The documentation should satisfy both academic and professional standards.

Objectives

Produce a complete documentation suite.

Prepare the final MSc report.

Create developer documentation.

Create deployment documentation.

Create user documentation.

Provide evidence supporting every assessment criterion.

Documentation Philosophy

Documentation is part of the product.

Explain decisions.

Do not repeat source code.

Support future developers.

Support assessors.

Support deployment.

Every document should answer:

What?

Why?

How?

Result?

Future Improvement?

Required Documents

Produce

Software Requirements Specification

System Architecture Document

Database Design Document

API Documentation

Security Documentation

Testing Documentation

Formal Methods Documentation

Deployment Guide

Developer Guide

Administrator Guide

User Guide

Maintenance Guide

Final MSc Report

Presentation Notes

README

Software Requirements Specification

Include

Introduction

Business Objectives

Stakeholders

Functional Requirements

Non Functional Requirements

Security Requirements

Privacy Requirements

Business Rules

Acceptance Criteria

Traceability Matrix

Project Scope

Glossary

Architecture Documentation

Explain

Architecture Overview

Technology Stack

Layered Architecture

Module Structure

Folder Structure

Data Flow

Authentication Flow

Authorization Flow

Deployment Architecture

Architecture Decision Records

Justify every architectural decision.

Database Documentation

Include

ER Diagram

Relational Schema

Prisma Schema

Relationships

Normalization

Constraints

Indexes

Migration Strategy

Seed Data

Backup Strategy

Restore Strategy

Data Dictionary

API Documentation

Document every endpoint.

Include

Purpose

Method

URL

Authentication

Authorization

Validation

Request

Response

Status Codes

Errors

Example Requests

Example Responses

Security Notes

Security Documentation

Document

Threat Model

OWASP Mapping

NIST Mapping

Authentication

Authorization

RBAC

Validation

Session Management

Rate Limiting

Audit Logging

Secrets Management

Residual Risks

Security Review

Every control should include its purpose.

Testing Documentation

Produce

Testing Strategy

Unit Test Report

Integration Test Report

API Test Report

Playwright Report

Performance Report

Accessibility Report

OWASP ZAP Report

SonarQube Report

Regression Report

Coverage Report

Bug Summary

Quality Dashboard

Formal Methods Documentation

Include

Finite State Machines

Petri Nets

Transition Tables

Reachability Analysis

Deadlock Analysis

Safety Verification

Liveness Verification

Formal Conclusions

Traceability

Every diagram should include interpretation.

Deployment Guide

Document

Prerequisites

Installation

Environment Variables

Supabase Setup

Prisma Setup

Docker

Docker Compose

Build Process

Deployment

Production Configuration

Backup

Recovery

Troubleshooting

Developer Guide

Explain

Project Structure

Architecture

Coding Standards

Adding Features

Creating APIs

Database Migrations

Testing

Deployment

Contribution Workflow

Administrator Guide

Explain

User Management

Role Management

Attendance

Performance

Medical Records

Reports

Audit Logs

Settings

System Maintenance

Backup

Recovery

User Guide

Create guides for

Administrator

Coach

Parent

Artist

Include

Login

Navigation

Daily Tasks

Common Workflows

Profile Management

Troubleshooting

Frequently Asked Questions

README

Include

Project Overview

Features

Technology Stack

Architecture

Installation

Configuration

Environment Variables

Running Locally

Testing

Deployment

Folder Structure

Screenshots

License

Contributors

Final MSc Report

Target approximately 2,000 words.

Structure

1. Introduction
2. Requirements Analysis

3. System Design
4. Secure Development Methodology
5. Implementation
6. Security Mechanisms
7. Testing and Analysis
8. Formal Methods
9. Legal and Ethical Considerations
10. Evaluation
11. Future Improvements
12. Conclusion
13. Harvard References

Use figures and tables throughout the report.

Diagrams

Produce

Use Case Diagram

Class Diagram

Sequence Diagram

Activity Diagram

ER Diagram

Deployment Diagram

Architecture Diagram

Component Diagram

FSM

Petri Net

Trust Boundary Diagram

Data Flow Diagram

Every diagram requires

Caption

Number

Explanation

Reference in text

Referencing

Use Harvard style.

Prefer

OWASP

NIST

Official Next.js Documentation

Official Supabase Documentation

Official Prisma Documentation

Peer Reviewed Journals

Academic Books

Avoid unreliable sources.

Deliverables

SRS

Architecture Document

Database Document

API Documentation

Security Guide

Testing Guide

Formal Methods Guide

Deployment Guide

Developer Guide

Administrator Guide

User Guide

README

Presentation Material

Final MSc Report

Reference List

Appendices

Acceptance Criteria

All documentation is complete.

Every feature is documented.

Architecture matches implementation.

Security decisions are justified.

Testing evidence is included.

Formal methods are documented.

Harvard referencing is consistent.

The report satisfies MSc academic standards.

Update Project Memory

Mark

Documentation

Completed

Record

Documents Produced

Report Version

Diagram Count

Reference Count

Outstanding Revisions

Next Task

Final Engineering Review

Documentation Review Checklist

Verify

Requirements are documented.

Architecture is explained.

Database is documented.

APIs are documented.

Security is justified.

Testing evidence exists.

Formal methods are complete.

Deployment guide is complete.

README is complete.

Report is professionally formatted.

Harvard references are correct.

Definition of Done

The project documentation is complete, accurate and professionally presented.

Every engineering decision is supported by clear explanations and evidence.

The final MSc report demonstrates critical analysis, secure software engineering practices, comprehensive testing, formal verification and professional technical communication, providing a complete submission suitable for distinction-level assessment.